Climate Informatics 2026 (2026), xx: 1–3
doi:10.1017/xxxx

CAMBRIDGE UNIVERSITY PRESS

CLIMATE INFORMATICS 2026
EXTENDED ABSTRACT

Diffusion Models for Generative Emulation of Regional Climate Models: Simulating Downscaling Uncertainty

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Received xx xxx xxxx

Keywords: diffusion model, emulator, regional climate model, uncertainty, generative machine learning

1. Introduction and objectives

By employing machine learning, *emulators* aim to provide a cost-effective alternative to regional climate models (RCMs). They do so by statistically replicating their dynamical downscaling function, therefore constituting a so-called *hybrid* methodology. Emulators work by empirically linking a set of large-scale predictors, well simulated by global climate models (GCMs), to RCM-simulated high-resolution fields of the variable of interest, such as precipitation or temperature. There is an inherent natural uncertainty in this downscaling function, both coming from the increase of resolution and from the predictors' limited information, specially for complex and highly local variables such as precipitation, the focus of this work.

In order to fully capture the downscaling function while taking into account this uncertainty, deterministic ML methods have been proven not enough (see recent studies such as [Doury (2024)]): they are unable to properly simulate the tail of the distribution and lack spatial detail when reproducing high resolution fields, especially over orographically complex regions. This makes generative machine learning models a sensible alternative to tackle the problem, which has made them gain much attention in recent years [Addison (2025), Aich (2024), Tomasi (2025)].

Among them, diffusion models are a generative machine-learning framework that learn the probability distribution of the data by progressively corrupting samples with gaussian noise. A deep learning model (typically a UNET [Ronneberger (2015)], called *backbone*) is trained to learn to reverse this process, and can be subsequently used for the generation of new samples via iterative denoising. They can be conditioned on large-scale predictors to act as RCM-emulators and are appealing for downscaling: they produce highly spatially-detailed samples that represent probability distributions rather than single deterministic fields, thus enabling large ensembles of downscaled realizations at substantially lower costs than those required to run an RCM. However, they are still more computationally demanding than state-of-the-art deterministic machine learning approaches, and the extent of their added value when downscaling GCM simulations still remains an open question. In addition, they exhibit more instability (for instance, diffusion models are known to produce *hallucinations*, see e.g., [Aithal (2025)]), potentially introducing additional biases into the modeling chain, and properly validating them is challenging, because of their stochastic nature and cost.

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Our objective is to perform a comprehensive and in-depth validation and comparison of diffusion-based modeling approaches for downscaling and emulation. To do so, we will move beyond the usual performance metrics currently used, such as root mean squared error or power spectral density, and focus on key challenges within downscaling and climate modeling, such as spatial extremes. We will also discuss practical guidelines (such as conditioning variables and evaluation metrics) and outline priorities for the generative climate downscaling community.

2. Methods and data

We address the question by employing a validation framework that allows us to perform the intensive tests needed. In particular, we are based on the experimental framework described in [Doury (2024)], working under the "perfect model" framework: we train our models with upscaled RCM variables and test them on different emissions scenarios. We emulate daily Precipitation at 0.11^o from the Aladin 63 RCM driven by CNRM-CM5 (EURO-CORDEX), over a 64×64 domain covering southwest France and northeast Spain. This is a challenging domain encompassing mountain ranges (The Cévennes and Pyrenees) and both the Mediterranean and Atlantic coasts.

We test different noise schedulers and conditioning approaches for diffusion models, assessing their impact or lack thereof in the final results. These results are in turn based on tests performed on specific extreme event targets. These are impossible to predict using deterministic models, so we assess whether the uncertainty envelope covered by the different diffusion model simulations can encompass them. We use probabilistic scores like the CRPS scores, while still checking if these generative models keep up with the good predictive performance observed in deterministic models.

3. Preliminary results and expected outcomes

Preliminary results indicate that diffusion models, no matter the approach, perform very well in all climatological aspects, producing highly detailed simulations that visually resemble the reference precipitation fields. They also properly generate, over long periods, precipitation amounts that are statistically very similar to the reference distribution, including the spatial tail. However, our early findings suggest that models are still unable to produce probability distributions with sufficient flexibility to adequately capture specific extremely heavy precipitation fields.

Acknowledgments. This study has received funding from *Agence Nationale de la Recherche - France 2030* as part of the PEPR TRACCS programme under grant numbers ANR-22-EXTR-0005 (EXTENDING project) and ANR-22-EXTR-0011 (LOCAL-ISING project). This project was provided with computing and storage resources by GENCI at IDRIS on the supercomputer Jean Zay's V100 partition.

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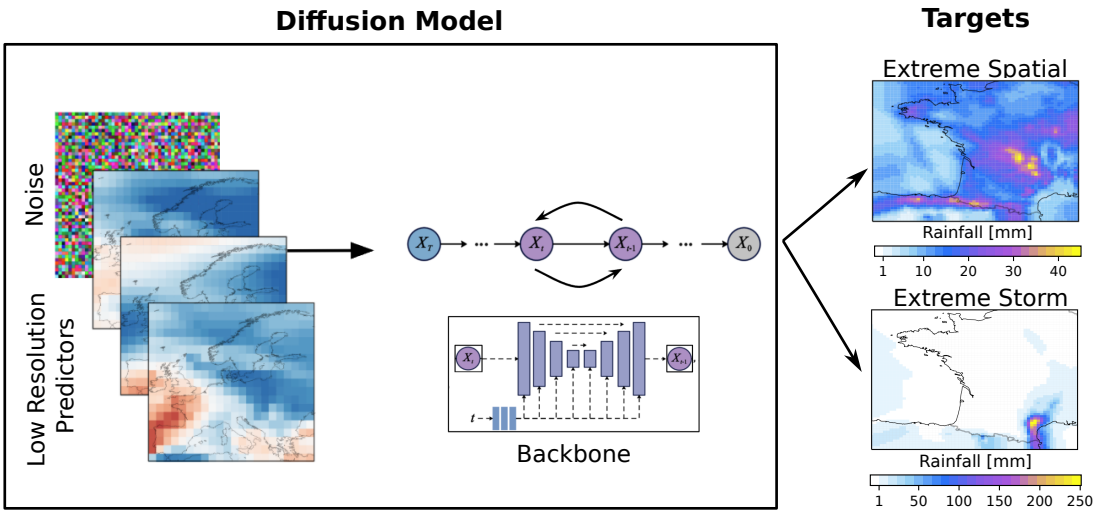

Figure 1. *Schematic that visually summarizes the research. We assess several diffusion model approaches and compare how they reproduce specific targets of interest..*

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